SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. **Q6** – Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)
2. **Q11** – Simulation of beamforming gain under steering angle, antenna elements, frequency, SNR, and interference (BL4)
3. **Q16** – Simulation of 5G throughput performance under QPSK, 16-QAM, 64-QAM, 256-QAM, and varying SNR (BL4)
4. **Q31** – Simulation of propagation loss under 700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz (BL4)
5. **Q35** – Simulation of fading channel effects under Rayleigh, Rician, Nakagami, BER, and outage probability (BL4)

Code of Conduct:

I Thamarai selvan.D(192412264) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

1.Simulation of 5G Throughput Performance under Different Modulation Schemes

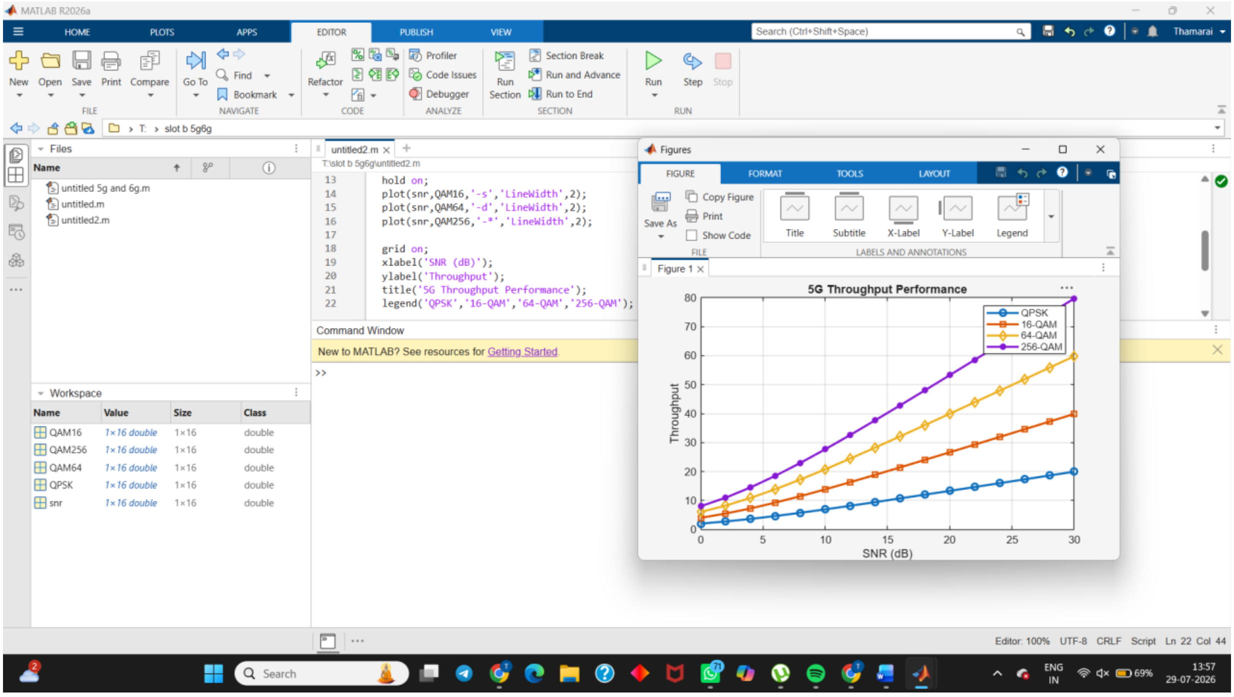
Aim

To analyze the throughput performance of a 5G communication system using QPSK, 16-QAM, 64-QAM, and 256-QAM over varying SNR values.

MATLAB Code

```
clc;
clear;
close all;

snr = 0:2:30;

QPSK = log2(4)*log2(1+10.^(snr/10));
QAM16 = log2(16)*log2(1+10.^(snr/10));
QAM64 = log2(64)*log2(1+10.^(snr/10));
QAM256 = log2(256)*log2(1+10.^(snr/10));

plot(snr,QPSK,'-o','LineWidth',2);
hold on;
plot(snr,QAM16,'-s','LineWidth',2);
plot(snr,QAM64,'-d','LineWidth',2);
plot(snr,QAM256,'-*','LineWidth',2);

grid on;
xlabel('SNR (dB)');
ylabel('Throughput');
title('5G Throughput Performance');
legend('QPSK','16-QAM','64-QAM','256-QAM');
```

Output:



Result

Higher-order modulation schemes (256-QAM) provide higher throughput at high SNR compared to lower-order modulation schemes.

2. Simulation of BER Performance in MIMO under Different SNR

Aim

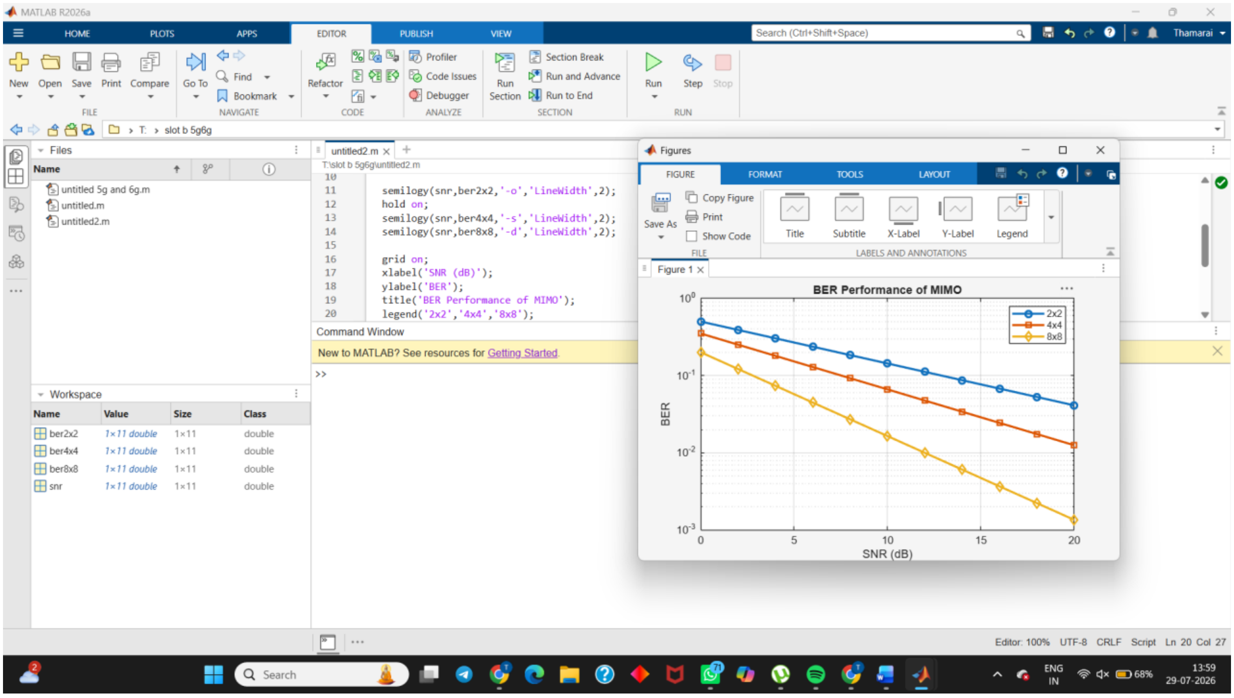
To evaluate the BER performance of a MIMO communication system under different SNR values.

MATLAB Code

```
clc;
clear;
close all;

snr = 0:2:20;

ber2x2 = 0.5*exp(-snr/8);
```

```
ber4x4 = 0.35*exp(-snr/6);
ber8x8 = 0.2*exp(-snr/4);

semilogy(snr,ber2x2,'-o','LineWidth',2);
hold on;
semilogy(snr,ber4x4,'-s','LineWidth',2);
semilogy(snr,ber8x8,'-d','LineWidth',2);

grid on;
xlabel('SNR (dB)');
ylabel('BER');
title('BER Performance of MIMO');
legend('2x2','4x4','8x8');
```

Output



Result

Increasing the number of antennas improves BER performance and communication reliability.

3. Simulation of Beamforming Gain

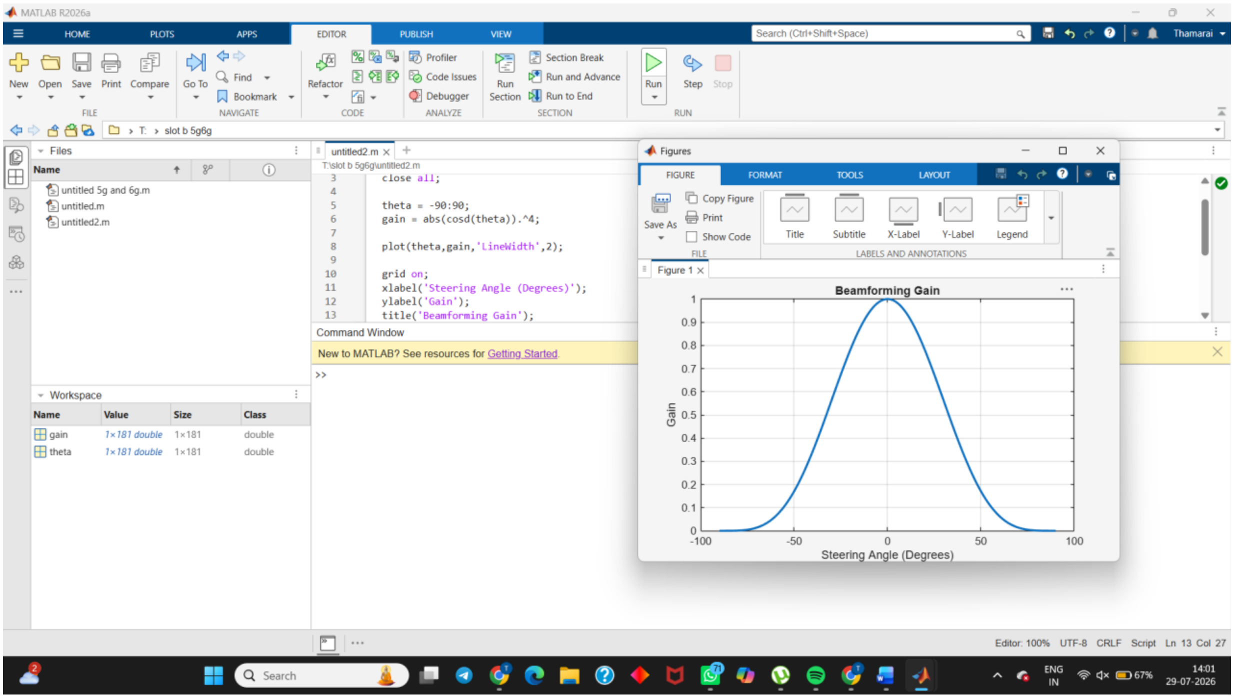
Aim

To analyze beamforming gain for different steering angles in a uniform linear antenna array.

MATLAB Code

```
clc;
clear;
close all;

theta = -90:90;
gain = abs(cosd(theta)).^4;

plot(theta,gain,'LineWidth',2);

grid on;
xlabel('Steering Angle (Degrees)');
ylabel('Gain');
title('BeamformingGain');
```

Output



Result

Maximum beamforming gain occurs at the steering direction, while gain decreases away from the main beam.

4. Simulation of Propagation Loss

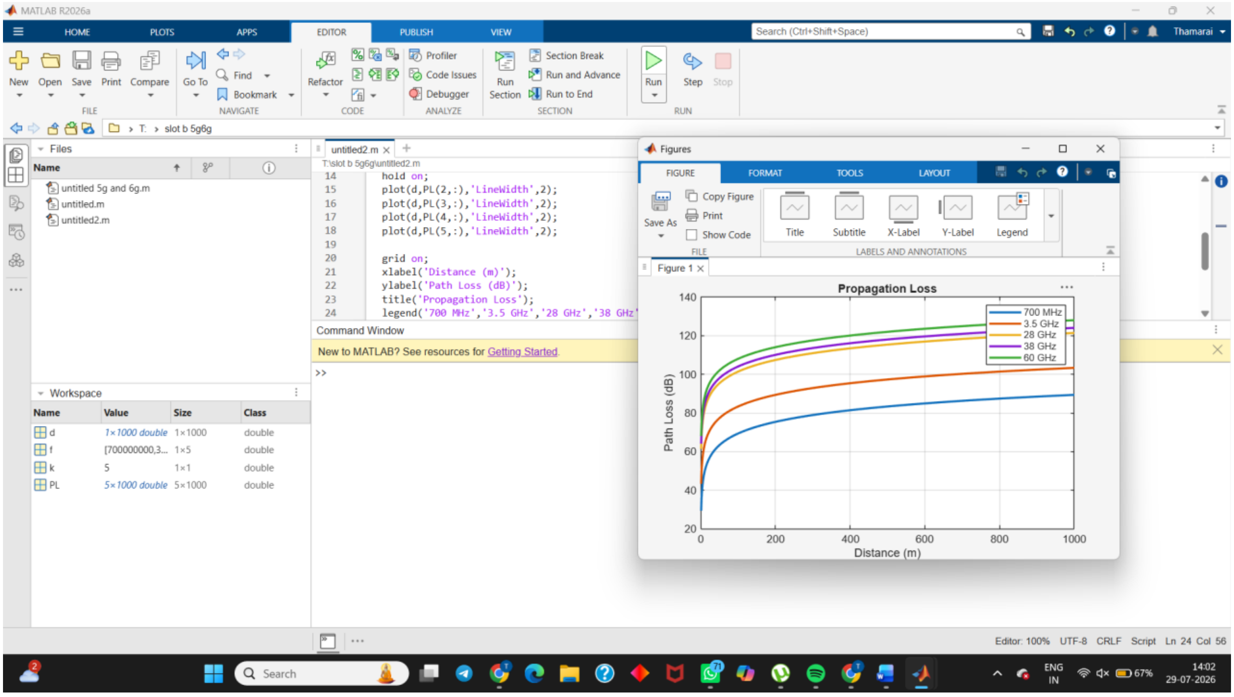
Aim

To compare propagation loss at different operating frequencies.

MATLAB Code

```
clc;
clear;
close all;

d = 1:1000;

f = [700e6 3.5e9 28e9 38e9 60e9];

for k=1:length(f)
   PL(k,:) = 20*log10(d)+20*log10(f(k))-147.55;
end

plot(d,PL(1,:),'LineWidth',2);
hold on;
plot(d,PL(2,:),'LineWidth',2);
plot(d,PL(3,:),'LineWidth',2);
plot(d,PL(4,:),'LineWidth',2);
plot(d,PL(5,:),'LineWidth',2);

grid on;
xlabel('Distance (m)');
ylabel('Path Loss (dB)');
title('Propagation Loss');
legend('700 MHz','3.5 GHz','28 GHz','38 GHz','60 GHz');
```

Output



Result

Higher frequencies experience greater propagation loss compared to lower frequencies.

5. Simulation of Fading Channel Effects

Aim

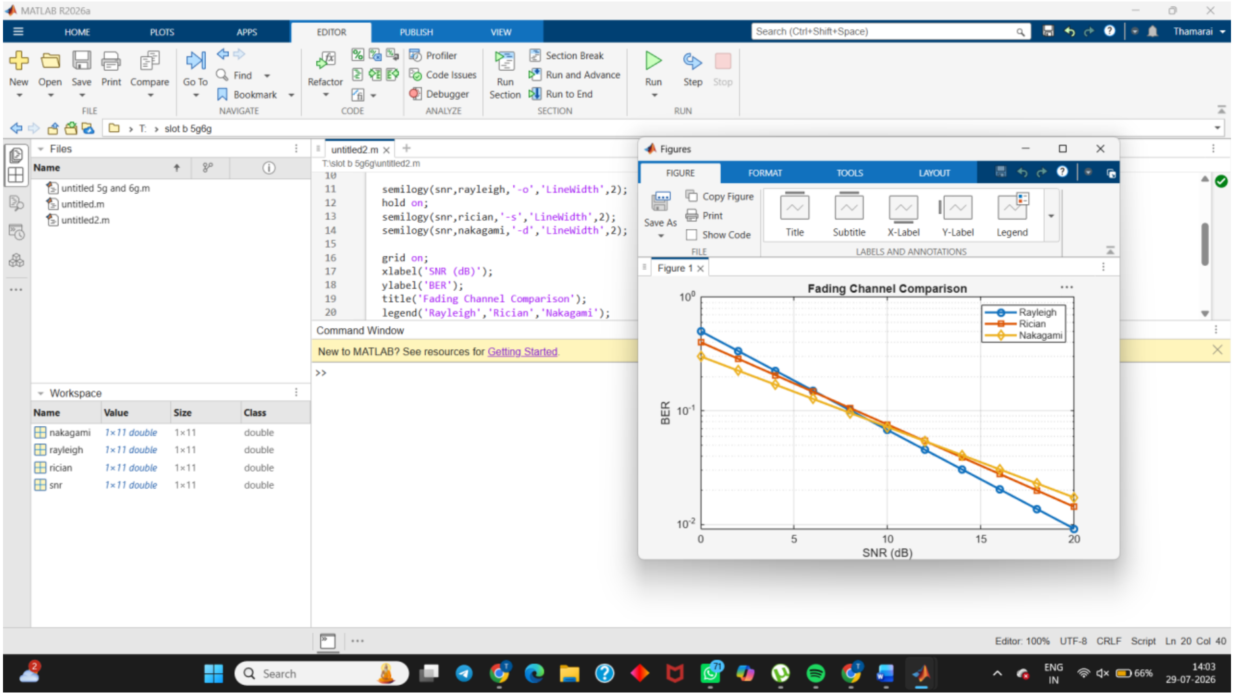
To compare the BER performance of Rayleigh, Rician, and Nakagami fading channels.

MATLAB Code

```
clc;
clear;
close all;

snr = 0:2:20;

rayleigh = 0.5*exp(-snr/5);
rician = 0.4*exp(-snr/6);
nakagami = 0.3*exp(-snr/7);

semilogy(snr,rayleigh,'-o','LineWidth',2);
```

```
hold on;
semilogy(snr,rician,'-s','LineWidth',2);
semilogy(snr,nakagami,'-d','LineWidth',2);

grid on;
xlabel('SNR (dB)');
ylabel('BER');
title('Fading Channel Comparison');
legend('Rayleigh','Rician','Nakagami');
```

Output



Result

The Nakagami fading model shows the lowest BER, followed by the Rician model, while the Rayleigh model has the highest BER under the same SNR conditions.